

Liang Haoran

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Target: Data Analyst | Locations: Hangzhou, Chengdu

PROFESSIONAL SUMMARY

Master's student in Business Analytics with a combined background in economics, statistics and data analysis, targeting data analyst roles. Proficient in SQL for data extraction, multi-table joins, group aggregation, CTEs, subqueries, window functions and business metric calculation; skilled in Python/Pandas for data cleaning, metric analysis and visualization, with familiarity in machine learning modeling, regression analysis, classification models and model evaluation. Participated in the full-stack development of the ARDashboard accounts-receivable management dashboard at Maxcess (a foreign company), gaining hands-on experience in ERP data extraction, MySQL data modeling and FastAPI/Vue dashboard delivery, and able to use vibecoding to assist SQL, ETL scripts and frontend/backend development; completed a data analysis internship related to Baidu search behavior data, covering data cleaning, conversion funnel analysis and user segmentation across the search-click-browse journey; independently completed the MavenFuzzyFactory e-commerce database project, with experience in traffic channel analysis, A/B testing, page conversion analysis and business growth analysis.

EDUCATION

University of Macau | MSc in Business Analytics | Faculty of Business Administration | Macau 2025.08-2027.07

· GPA: 3.7/4.0, top 5% of the program; Core courses: Data Visualization, Data Science, Machine Learning, Database & Big Data Management (SQL), Statistical Learning and Data Mining, Python

Yunnan University of Finance and Economics | BSc in Trade Economics | School of Economics | Kunming 2021.09-2025.06

· GPA: 3.48/4.0, top 10% of the program; First-Class University Scholarship for consecutive years; Major coursework: Economics, Statistics, Advanced Mathematics, Linear Algebra, Econometrics

INTERNSHIP EXPERIENCE

Maxcess | IT Data Development Intern 2026.05-2026.09

- Requirement analysis & metric alignment: worked with Finance, Supply Chain and IT to define core metrics for AR management including aging buckets, overdue determination, prepayment write-off and DSO; consolidated AR, receipt, shipment and general-ledger data scattered across the Kingdee K3 ERP system into an analyzable MySQL data model, and produced standardized metric-definition documentation.
- Data modeling & ETL development: designed 18 business tables and 3 public views with a composite unique key of "AR bill no. + period + ledger" for deduplication; wrote Python extraction scripts to periodically sync ERP data into MySQL, accumulating 23K AR detail rows, 1.09M shipment rows and 18K receipt/write-off records, covering 5 settlement organizations, 1,200+ customers, 23 sales reps and 4 currencies.
- AI-assisted full-stack development: based on existing backend templates, used vibecoding approaches such as CodeBuddy and Codex to develop and tune 27 FastAPI+SQLAlchemy backend APIs and 4 Vue3+ECharts frontend pages, implementing multi-table joins, multi-currency USD conversion and automatic 7-bucket aging, and reviewed the metrics and boundary conditions of AI-generated SQL line by line.
- Visualization & delivery: deployed via Nginx to the corporate intranet for daily use by Finance and Sales, replacing the previous manual Excel reporting process; a single aging and overdue report dropped from ~4 hours to within 10 minutes, with overdue customers, collection commitments and prepayment write-off status visually tracked.

Baidu | Data Analyst Intern 2026.03-2026.04

- Business analysis positioning: focused on Baidu's core search engine business chain; used SQL, Python and Excel to conduct dedicated analysis of user search behavior, traffic channel distribution and the search-click-browse conversion funnel, identifying core influencing factors such as search result ranking, keyword type, traffic source and device type, and locating existing problems and 2 key conversion drop-off points.
- Data cleaning: queried and filtered user search behavior data with SQL, independently processing 100K user search behavior log rows covering 18 fields such as user ID, search time, device type, traffic source, keyword type, search result ranking, click behavior and browse behavior; used Python Pandas to perform MCAR tests on missing values and field-definition consistency checks, identified a definitional contradiction in the result_rank field, and completed outlier handling, field standardization and feature derivation to build a standardized analysis dataset.
- User segmentation analysis: performed user segmentation and behavioral profiling with Python, breaking down click and browse performance across 4 traffic channels, 4 keyword types, 2 device types and 5 city tiers; built the full search-click-

browse conversion funnel, quantifying a ~30.0% drop-off from search to click and ~9.8% from click to browse, and used Pandas/SQL for multi-dimensional metric statistics and business attribution analysis.

- Visualization: created 12 professional charts with Matplotlib and Seaborn covering traffic source distribution, device-side differences, keyword-type CTR, search time trends and user browse conversion funnels; delivered 1 interim analysis report together with tracking-instrumentation improvement recommendations to support search journey optimization, traffic channel evaluation and subsequent strategy adjustments.

Industrial Bank Taizhou Branch | Data Analyst Intern

2025.06-2025.09

- Credit risk data analysis support: participated in corporate credit risk data analysis, observing and assisting data analysts in extracting corporate customer financial, transaction-flow and basic information data from business databases; understood the construction logic of risk indicators such as profitability, solvency and operational stability, and helped organize foundational customer risk analysis data tables.
- Data cleaning & quality checks: assisted with SQL/Excel completeness and consistency checks on customer basic information, financial indicators and transaction data, identifying missing values, duplicate records and abnormal fields, and compiling data quality issue lists to provide a more standardized data foundation for post-loan management and risk analysis.
- Reporting & business support: participated in monthly departmental reporting, helped maintain standardized Excel filing templates, and used pivot tables, basic functions and data validation to improve data entry and checking efficiency; followed up on relationship managers' post-loan management data needs and understood how data analysis supports customer risk identification and business decisions.

PROJECT EXPERIENCE

ARDashboard Accounts-Receiveable Management Dashboard | Corporate Internship Project

2026.06-2026.09

- Background: AR management relied on Finance manually exporting multiple Excel sheets from Kingdee K3 each month to compile aging and overdue reports, with inconsistent metrics and lagging updates, and no visual tracking of overdue collections or prepayment write-offs; the project aimed to connect the AR, receipt, shipment and general-ledger data chain and build an integrated dashboard covering aging structure, overdue distribution and DSO trends.
- Data modeling & sync: designed 18 MySQL tables (AR details, unbilled items, receipt bills, write-off records, account balances, exchange rates) and 3 views, defining unique keys and incremental sync rules; wrote Python extraction scripts to connect to Kingdee K3 for field mapping, deduplication and incremental loading, supporting stable queries of millions of detail rows spanning 2013-2026, and extended a separate refund database linking refund bills to their source receipt bills.
- Metric calculation & API development: used SQL multi-table joins, CASE WHEN grouping, date functions and aggregations to implement automatic 7-bucket aging (Uninvoiced / Not Due / Due / Overdue 1-30 / 31-60 / 61-90 / 91+), multi-currency local-currency and USD conversion, organization-sales-customer three-level aggregation and DSO calculation; delivered 27 FastAPI query APIs supporting drill-down from customer summaries to individual AR bill details.
- Visualization & delivery validation: used Vue3+Element Plus+ECharts to develop KPI cards, currency-grouped summary tables with automatically injected subtotal and total rows, a customer detail drawer and DSO/aging trend charts, covering 5 settlement organizations and 1,200+ customers; used AI coding assistants to generate and optimize SQL and frontend components, and after manual metric validation deployed to the intranet, improving report generation from ~4 hours to within 10 minutes.

MavenFuzzyFactory E-commerce User Behavior & Conversion Analysis Project | Course Project

2026.01-2026.05

- Background: conducted SQL data analysis on the Maven Fuzzy Factory e-commerce database, covering website traffic, user behavior paths, page conversion efficiency and marketing channel performance across core business tables including website_sessions, website_pageviews, orders, order_items, products and order_item_refunds.
- Traffic channel analysis: used SQL multi-table joins, GROUP BY, CASE WHEN and aggregate functions to break down session, order and CVR performance across gsearch, bsearch, organic search and direct type-in channels; further analyzed channel quality by brand/nonbrand, device type and utm_source to inform advertising budget allocation and keyword bidding strategy.
- Page conversion & A/B testing: conducted A/B testing analysis on the /home and /lander-1 landing pages, using temporary tables/CTEs to build landing page session datasets and calculate total sessions, bounced sessions and bounce rate; results showed the new landing page had a lower bounce rate than the homepage, supporting page optimization decisions.
- User behavior funnel & business growth: built a full-funnel conversion path from homepage/landing page through product page, product detail page, cart, shipping and billing to payment success based on website_pageviews, locating key drop-off nodes; extracted sessions, orders and CVR by month/week to analyze business trends across branded/non-branded keywords and mobile/desktop users and interpret the results.

- Data & feature engineering: built a click behavior analysis framework based on search behavior logs; used Python/Pandas to distinguish structural missingness (naturally empty when the event did not occur), such as click time and browse duration, from true missingness, filling only fields such as result ranking and keyword length with medians plus missing flags while handling abnormal records; constructed feature variables including search result ranking, keyword type, device type, traffic source, time slot and time to first click, and defined whether the user clicked as the target variable.
- Model analysis & business conclusions: first performed feature screening with univariate AUC and mutual information, then cross-validated the impact of different features on click behavior using binary logistic regression and gradient boosting tree models, and used a shuffled-label control group to rule out pipeline leakage; evaluated model performance with accuracy, recall, AUC, Wilson confidence intervals and multiple-comparison correction; summarized the core factors affecting CTR to provide a data analysis basis for search result ranking optimization, keyword strategy adjustment and recommendation material optimization.

HONORS & AWARDS

- National College Students Statistical Modeling Competition, Provincial Third Prize | 2024: applied machine learning algorithms such as XGBoost to gold data, completing data preprocessing, feature construction, model training and result analysis.
- MCM (Mathematical Contest in Modeling), Successful Participant Award | 2023: used R for factor extraction, regression analysis and modeling, gaining experience in statistical analysis and modeling competitions.

SKILLS

- Data analysis: proficient in SQL for data extraction, table joins, group aggregation, window analysis, CTEs, subqueries and business metric calculation; familiar with Excel pivot tables, basic functions and data cleaning.
- Programming & modeling: proficient in the Python data analysis workflow, able to use AI to work with Pandas, NumPy and Matplotlib for data cleaning, metric calculation, visualization and project analysis; familiar with statistical analysis and machine learning methods, covering regression analysis, classification models, feature engineering and model evaluation.
- Data development: hands-on experience in MySQL data modeling and ETL extraction (table structure design, unique-key deduplication, incremental sync); familiar with FastAPI+SQLAlchemy backend API development and Vue3+ECharts dashboard visualization; practical experience using AI coding tools (CodeBuddy, Codex) to assist SQL, ETL scripts and frontend/backend development.
- Others: basic Tableau; proficient in R for statistical modeling, factor analysis and regression analysis; IELTS 6.0, CET-6.

PORTFOLIO

- Online portfolio (projects / skills / resume): <https://lianghaoran1118-png.github.io/portfolio-site/>